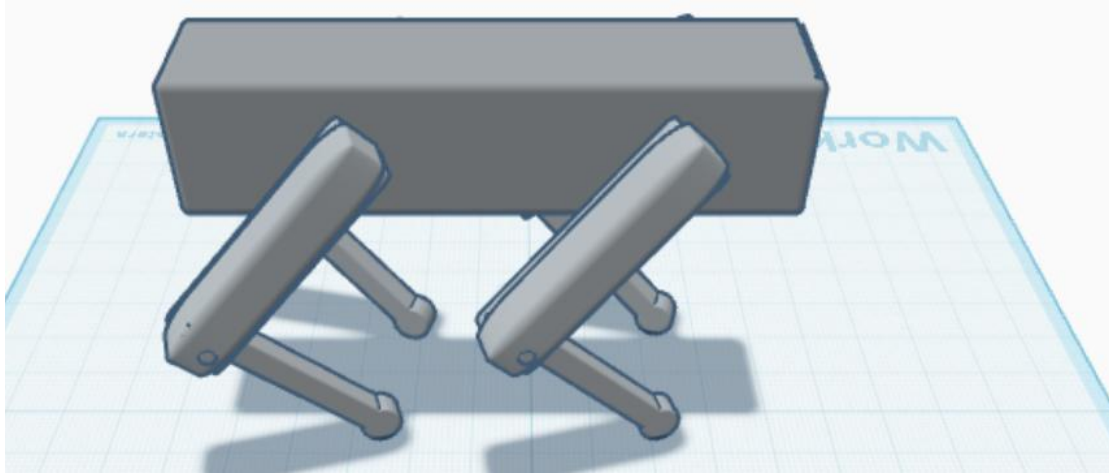


Mechanical Design of a Simple Robotic Dog

Body and Chassis Shape

The robot has a simple rectangular chassis that provides space for the battery, controller, and motors. The design is lightweight and easy to manufacture.



- Leg Design

Each leg consists of two links (upper and lower leg) connected by revolute joints. The rounded feet help improve ground contact and stability.

-Number of Joints and Degrees of Freedom (DOF)

Each leg has:

1 Hip joint AND 1 Knee joint

So , The Total Degrees of Freedom is $2 \times 4 = 8$

Motor Selection

The **MG996R servo motor** is selected because it provides sufficient torque, is easy to control, and is suitable for lightweight robotic projects.

Preliminary Torque Calculation

Assume:

Robot mass = **2 kg**

Leg length = **0.10 m**

Weight on one leg:

$$F = \frac{2 \times 9.81}{4} = 4.9 \text{ N}$$

Torque at the hip joint:

$$T = F \times L$$

$$T = 4.9 \times 0.10 = 0.49 \text{ N} \cdot \text{m}$$

With a safety factor of **2**:

$$\text{Required Torque} \approx 1.0 \text{ N} \cdot \text{m}$$

Therefore, the **MG996R servo motor** is suitable for this design.

Stability and Center of Gravity

The center of gravity is located near the middle of the body. Keeping the battery and heavy components in the center improves balance and stability during standing and walking.

Proposed Walking Gait

The robot uses a **Crawl Gait**, where one leg moves at a time while the other three legs remain on the ground. This provides stable and simple movement.

Expected Mechanical Problems

Insufficient motor torque if the robot is too heavy.

Joint wear after repeated use.

Foot slipping on smooth surfaces.

Balance problems if the weight is unevenly distributed.